# On-line Avoidance strategy with Multiple No-fly Zones

Lianjie Shen, Jianwen Zhu

*Abstract*—**A novel online avoidance strategy with multiple no-fly zones is proposed based on real-time selection and dynamic programming. The no-fly zones are classified and screened via relative position between flight path and the no-fly zones described by route offset distance, and the boundaries to flyover are determined by using the dynamic programming method. With the boundary points, the entire gliding flight process is divided into several series stages, and then the sub-guidance algorithm is employed to realize multiple constraints guidance. Finally, the simulation is carried out to show that this method can avoid more than one no-fly zone and provide a reference for further research on avoidance guidance in multi-no fly zone.**

## I. Introduction

Hypersonic gliders are receiving tremendous attention in the combat context of long-distance fleetly accurately strike. This type of aircraft, in certain special missions (such as the U.S. Air Force's global strike mission [1]), is required to circumvent some areas due to threats and geopolitical factors during flight. These areas are so called a no-fly zone.

In recent years, the issue of gliding trajectory planning and guidance considering the no-fly zone constraints has been paid attention to by scholars. Jorris and Cobb firstly studied two-dimensional and three-dimensional optimal trajectory generation methods[2][3]to meet the constraints of waypoints and no-fly zones for common aero vehicle (CAV). Based on the CAV model, Hu[4] put forward a geometric method to meet the no-fly zone lateral avoidance. Considered the lateral maneuvering ability need of passing the waypoint and avoiding no-fly zone, Xie[5] designed the drag acceleration profile to satisfy the waypoint constraints and to avoid the no-fly zone. Aimed at the problem of gliding reentry with no-fly circle, Yong [6] proposed a parameters coupled design method of the maneuvering trajectory and aerodynamic characteristics. To guide the vehicle avoid no-fly zone, Wang [7] transformed no-fly zone constraint into heading angle constraint in real time and applied the dynamic compensation strategy of heading deviation corridor to control the bank angle sign. The above trajectory planning and guidance methods were concentrated on the online avoidance considering a few no-fly zones and off-line trajectory planning considering multiple no-fly zones. In order to ensure real-time requirements, and guarantee local optimality at the same time, we have carried out the related study.

For the problem of multiple no-fly zones avoidance, a novel online no-fly zones avoidance strategy is proposed in this paper. First of all, the no-fly zones are divided into three categories according to their characteristics. Secondly, based on the dynamic programming method, the minimum route offset distance is taken as the decision basis, and the avoidance problems for multiple no-fly zones are converted into waypoints constraint. Thirdly, The flight path points obtained by this strategy divide the flight process into multiple phases, and then take the step-by-step guidance to avoid more than one no-fly zone with less energy consumption. Finally, the effectiveness of this strategy is verified by simulation.

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## II. Problem description and avoidance strategy

### A. Problem description

Many scholars have put forward their own methods for the guidance problem without no-fly zones.In contrast, for the no-fly zones, and when there are more no-fly zones, the guidance issue becomes relatively complicated. The idea of this paper is to transform the no-fly zone constraint into the waypoint constraint ( $h_{P_k}, \lambda_{P_k}, \phi_{P_k}$ ) by the dynamic programming method, and divide the entire gliding segment into multiple stages. In this way, the gliding segment guidance problem with no-fly zone is transformed into gliding parade guidance problem in non-no-fly zone. That is to say, given the initial conditions of the aircraft, guiding the aircraft no-impetus gliding to the set stages end-point (planned waypoint) ( $h_{P_k}, \lambda_{P_k}, \phi_{P_k}$ ) under the complicated process constraints.

### B. Avoidance strategy

Considering the extensiveness and complexity of the no-fly zone in the spatial distribution, according to the impact of the no-fly zone on the flight process to the target process, firstly, the no-fly zone should be screened as the first sort in the area with great influence; secondly, The first sort no-fly zones classified act as the steps of dynamic planning, and the route offset distance of the second-class no-fly zones can be calculated. Using the dynamic programming method, the minimum route offset total distance is used as the decision basis to make the hierarchical decision. By this way, waypoints are planned. Then, select the appropriate gliding guidance law to guide the aircraft to the online planning waypoint by segment guidance. Finally, the task of avoidance and reaching the target point is achieved. The specific process is shown in Figure 1.

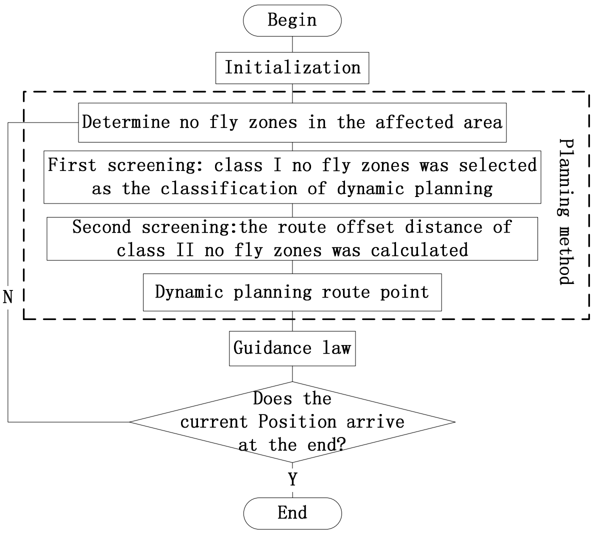

Figure 1. Avoidance Strategy Flowchart

## III. CLASSIFICATION AND SCREENING OF NO FLY ZONES

In view of the complexity of the space distribution in the no fly zone, the following definitions and classifications are carried out to realize the online planning for the rapid planning of the route points.

OT is got by connecting the starting point O to the target point T. Extending 3 Rnf_max to the direction perpendicular to OT on both sides (Rnf_max is the largest no fly zone radius in the vicinity area),a rectangular area would be formed . We call it the "no fly zone influence area"("influence area" for short). The center points' latitude and longitude of the no fly zones and the radius of the no fly zones are used as input parameters in this area, so as to be further screened and planned.

There may be some no fly zones in the direction of the starting point to the target point. The no fly zones hindered the flight from the starting point to the target point. These zones, which defined as "the first kind of no fly zones", must be avoided, such as the No.2 and No. 4 no fly zones in the Figure 2.For each of the first kind of no fly zone, there are two different flying directions. There may be some no fly zones in the two directions, which have a bad influence on the flying around the first kind of no fly zones. This kind of no fly zones are called "the second kind of no fly zone", such as the No.1 and No. 6 no fly zones in the Figure 2.The no fly zones in the affected area, which will not affect the avoidance flight of the first two types of no fly zones, are called the third kind of no fly zones, such as the No.3 and No. 5 no fly zones in the Figure 2.

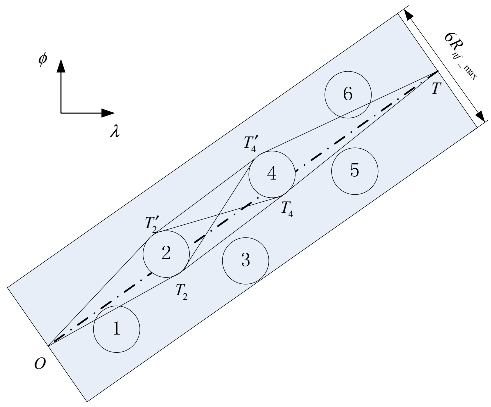

Figure 2 .The Affected Area of No-fly Zones Sketch Map

The first screening: as shown in Figure 3, O as the start point, $T$ as the goal, $T_1'$ as the above boundary point of no.1 no fly zone, the lower boundary point is $T_1$ . The above boundary point of no.2 no fly zone is $T_2'$ and $T_2$ is the lower one. $\sigma_{OT}$ , $\sigma_{OT_1}$ and $\sigma_{OT_1'}$ are the track angles of $OT$, $OT_1$ and $OT_1'$. They can be calculated by the spherical triangle theory :

$$\tan\sigma_{LOS} = \frac{\sin\left(\lambda_{nf\_k}-\lambda\right)}{\cos\phi\tan\phi_{nf\_k}-\sin\phi\cos\left(\lambda_{nf\_k}-\lambda\right)} \quad (1)$$

Where, $\lambda_{nf\_k}$ and $\phi_{nf\_k}$ are the latitude and longitude of the boundary point of the No. k no fly zone or target point. Obviously, when the no fly zone must be circumvent, for example, the No.1 no fly zone in Figure 3, $\sigma_{OT} \in \left[\sigma_{OT_1'}, \sigma_{OT_1}\right]$. On the contrary, the no fly zone may not need to be circumvent, such as the No.2 no fly zone in Figure 3. $\sigma_{OT_2}$ and $\sigma_{OT_2'}$ are the track angles of $OT_2$ and $OT_2'$.

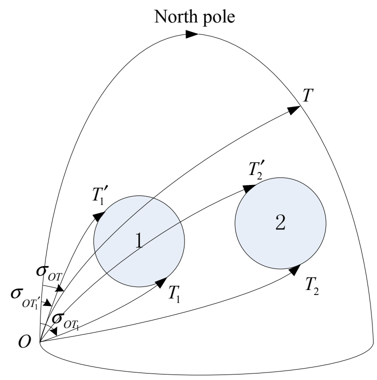

Figure 3 The First Screening

The second selection: unlike the first screening, the second screening is to find the no fly zones which are on the side of the class I no-fly zones when it is avoided. In Figure 2, two no fly zones, 2 and 4, which must be avoided, are obtained by the first screening. They divide the whole area into 3 segments. These 3 segments can be seen as 3 small affected area. The starting point is the O or the boundary point of the former no fly zone. The target point is the boundary point of the later no fly zone or the T. Then the first screening method was used to screen other no fly zones within the small affected area as a screening object. For example, there is a screening object No.3 in the small impact area between the No.2 and No.4 no fly zones, therefore the No.3 no fly zone does not need to be circumvented. In the same way, No.1 and No.6 can be screened for the possibility of avoiding flying zones, and No.3 and No.5 are no flight zones that need to be avoided.

For the case of Figure 2, the screening of all the no-fly zones can be done through the above two screening. However, there is also a case where these no-fly zones further divide the "small impact area" of the second screening into smaller ones after the second screening. If no-fly zones exist in these smaller areas, these no-fly zones may be class II no-fly zones. Therefore, the second screening process needs to be repeated several times until there is no second type in the small impact area.

In view of the limitations of aircraft control capabilities and the complexity of the avoidance issues, can be completely avoided by re-selecting the launch points that the case of two or more class II no-fly zones exist between the class I no-fly zones. As a result, the problem has been reduced to only one class II no-fly zone between the class I no-fly zones. In this case, only two screenings are required. After screening out, dynamic programming is used to get the waypoint, and then the segmented guidance. Because there is an error between the position of aircraft and the ideal position, it is necessary to plan the waypoint in real time.

## IV. DYNAMIC PLANNING WAYPOINTS

As shown in Figure 4, in the latitude and longitude plane, there are n class I no-fly zones from the starting point O to the target point T. For the *i*-th no-fly zone, let the center be $O_i$, the radius $R_i$. The latitude and longitude distance from $O_i$ to line segment OT is $\Delta l_i$ ($O_i$ is positive at the top left of *OT*, negative otherwise). The boundary points is $p_{i,1}$ and $p_{i,0}$ ($p_{i,1}$ $p_{i,0}$ perpendicular to *OT and Pass the point* $O_i$).

The no-fly zone avoidance will inevitably lead to energy loss, in order to reduce unnecessary losses, select the minimum energy loss as the optimal performance indicators:

$$\min J = \sum_{i=1}^{n} \|u_i\| \tag{2}$$

Among them, $u_i$ is the additional energy consumed by avoiding the *i*-th class I no-fly zone.

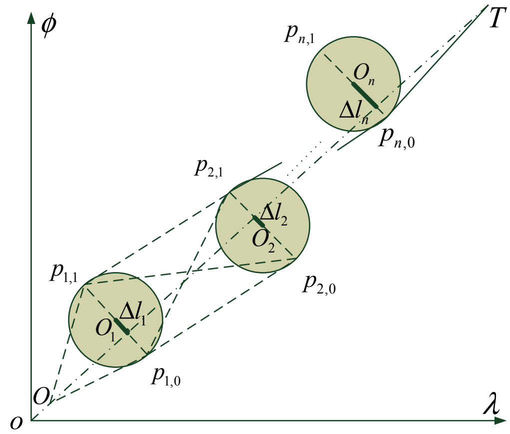


Figure 4. Optional Routes of Multiple Class I No-fly zones

However, the value is difficult to calculate. It is taken into account that the greater route offset distance the greater the energy required to be consumed. Therefore, it may be desirable to select the minimum route offset distance as the performance index, i.e.

$$\min J = \sum_{i=1}^{n} \|u_i\| \triangleq \sum_{i=1}^{n} d_i \tag{3}$$

Where, $d_i$ is the route offset distance to avoid the *i*-th class I no fly zone.

$$d_i = d'_i + d''_i \tag{4}$$

The $d_i$ consists of two parts. The first part is the route offset distance between the class I no fly zones. The other part is the route offset distance between the class I no fly zone and class II no fly zone. If there is no class II no fly zone between the class I no fly zones,

$$d''_i = 0 \tag{5}$$

As shown in Figure 5, when there are class II no fly zones between the stages, it is more sensible to fly around the small side. Save the boundary points of class II no fly zones that need to go through when flying around small side, and calculate the route offset distance which needs to be avoided by the following expression:

$$d''_i = R_j / R_e - \Delta l'_j \tag{6}$$

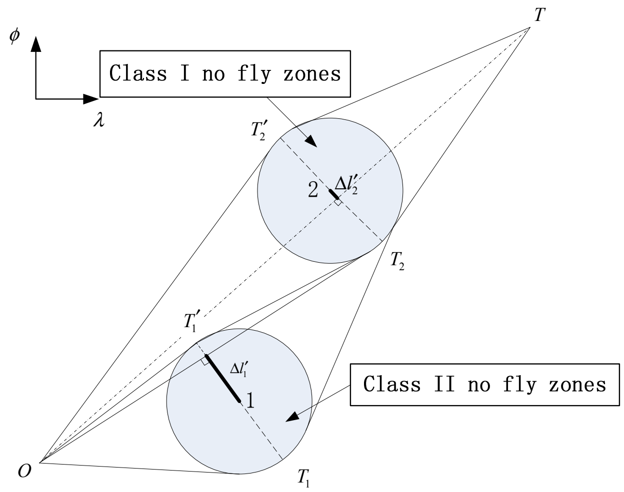


Figure 5. Route Offset Distance of Class II No Fly Zone

The process from $O$ to $T$, the route offset distance $d_i'$ required by the boundary points can also be solved through geometric relations. The solution will be detailed below.

The total route offset distance from the starting point O to the target point T is the least, which is actually a multi-stage decision process. The avoidance process of the whole no fly zone is divided into several stages, and each level must make decisions (determine the direction of flying around) so as to achieve the best effect of the whole process.

An optimal decision has such properties. No matter what the initial state and initial decision is, the remaining decisions must constitute an optimal decision for the state caused by the first decision, which is called the optimality principle. It can also be simply described as, "the second segment of the optimal trajectory, which is also the optimal trajectory[8]. "

When the dynamic programming method is used to solve the optimal decision, the multi-stage decision process is first expressed as the following mathematical expression [8] according to the optimality principle.

$$\omega_k\left(p_{n+1-k,i}\right)=\min_{j\in\{0,1\}}\left[d\left(p_{n+1-k,i},p_{n+2-k,j}\right)+\omega_{k-1}\left(p_{n+2-k,j}\right)\right] \quad (8)$$

Among them, $k=2,3,\ldots,n$, $i\in\{0,1\}$, $\omega_k\left(p_{n+1-k,i}\right)$ is the minimum total route offset distance from the beginning $p_{n+1-k,i}$ to the end $T$ of the $k$ stage decision process, and $d\left(p_{n+1-k,i},p_{n+2-k,j}\right)$ is the lateral displacement from the starting point $p_{n+1-k,i}$ to the higher level starting point $p_{n+2-k,j}$ of the $k$ stage decision process.

The equation (8) shows that in order to achieve the minimum total route offset distance in the decision process of the $k$ stage, the decision should be made in accordance with the principle of the minimum sum of the two lateral displacement. The first part is the one step route offset distance of the starting point to the higher level, and the second part is the total route offset distance of the higher level decision.

It is remarkable that the route offset distance mentioned in this paper is the distance defined in the plane of latitude and longitude, so its unit is radian.

The first step route offset distance is as follows:

$$\begin{cases} d'\left(p_{n,1},T\right)=R_n/R_e+\Delta l_n \\ d'\left(p_{n,0},T\right)=R_n/R_e-\Delta l_n \end{cases} \quad (7)$$

The one step route offset distance of the last stage is

$$\begin{cases} d'\left(O,p_{1,1}\right)=R_1/R_e+\Delta l_1 \\ d'\left(O,p_{1,0}\right)=R_1/R_e-\Delta l_1 \end{cases} \quad (8)$$

As the same, one step route offset distance can be obtained by the geometric relationship.

$$\begin{cases} d'\left(p_{n+1-k,1},p_{n+2-k,1}\right)=\left|R_{n+2-k}/R_e+\Delta l_{n+2-k}-R_{n+1-k}/R_e-\Delta l_{n+1-k}\right| \\ d'\left(p_{n+1-k,1},p_{n+2-k,0}\right)=R_{n+1-k}/R_e+\Delta l_{n+1-k}+R_{n+2-k}/R_e-\Delta l_{n+2-k} \\ d'\left(p_{n+1-k,0},p_{n+2-k,1}\right)=R_{n+1-k}/R_e-\Delta l_{n+1-k}+R_{n+2-k}/R_e+\Delta l_{n+2-k} \\ d'\left(p_{n+1-k,0},p_{n+2-k,0}\right)=\left|R_{n+2-k}/R_e+\Delta l_{n+1-k}-R_{n+1-k}/R_e-\Delta l_{n+2-k}\right| \end{cases} \quad (11)$$

The latitude and longitude distance from the $O_i$ to the line segment $OT$ is

$$\Delta l_i=\begin{cases} \dfrac{\left|k_{OT}\cdot\lambda_{nf\_k}-\phi_{nf\_k}+b_{OT}\right|}{\sqrt{k_{OT}^2+1}} & \left(k_{nf\_k}>k_{OT}\right) \\ -\dfrac{\left|k_{OT}\cdot\lambda_{nf\_k}-\phi_{nf\_k}+b_{OT}\right|}{\sqrt{k_{OT}^2+1}} & \left(k_{nf\_k}\le k_{OT}\right) \end{cases} \quad (9)$$

Among them, $b_{OT}=\phi_O-k_{OT}\cdot\lambda_O$, $k_{nf\_k}=\dfrac{\phi_{nf\_k}-\phi_O}{\lambda_{nf\_k}-\lambda_O}$ .For the first stage, the minimum route offset distance of the present point to the end is a one step route offset distance. When the equation (9)~(13) is put into (8), the former $n$ stage decision can be made. The value of $j$ is determined by the decision. When $j=1$, the current point is connected to the north point of the next stage; When $j=0$, the current point was connected to the south point of the next stage. In this way, two optimal decisions for each level are obtained. Two routes can be obtained through the previous $n$ level decision, and the level $n$+1 decision (final stage) determines which one is the best line.

$$\omega_{n+1}(O)=\min_{j\in\{0,1\}}\left[d\left(O,p_{1,j}\right)+\omega_k\left(p_{1,j}\right)\right] \quad (10)$$

When the longitude and latitude of the No. k no fly circle $\left(\lambda_{nf\_k}, \phi_{nf\_k}\right)$ and the radius of the no fly circle $R_k$ are known, the latitude and longitude of the waypoint $p_{k,1}$ can be solved by plane geometry.

$$\begin{cases} \lambda_{P_{k,1}} = \lambda_{nf\_k} - \dfrac{R_k}{R_e} \cdot \dfrac{k_{OT}}{\sqrt{k_{OT}^2 + 1}} \\ \phi_{P_{k,1}} = \phi_{nf\_k} + \dfrac{R_k}{R_e} \cdot \dfrac{1}{\sqrt{k_{OT}^2 + 1}} \end{cases} \tag{11}$$

The longitude and latitude of $p_{k,0}$ can also be obtained. Where the $k_{OT}$ is the slope of the starting point $O$ to the target point $T$ connection

$$k_{OT} = \frac{\phi_T - \phi_O}{\lambda_T - \lambda_O} \tag{12}$$

In this way, the constraint of the no fly zone is transformed into the route point constraint by the dynamic programming method. The problem of multiple no-fly zones avoidance is also transformed into the problem of one by one route point guidance.

## V. Comparative Analysis of Algorithm

In order to better understand the algorithm proposed in this paper, the following is the comparison and analysis between the traditional multiple no fly zones avoidance planning algorithm and the algorithm proposed in this paper.

(1) The purpose is the same, to avoid multiple no fly zones.

(2) The output is different. The output of the traditional algorithm is the planned trajectory, but the output of the output of this algorithm is the boundary points.

(3) Different solutions: The traditional algorithm is realized through trajectory planning and trajectory tracking. The algorithm of this paper is achieved by obtaining boundary points and subsection guidance.

(4) The comparison between algorithm complexity and real time is shown below.

At present, there are A* algorithms, genetic algorithms and other intelligent optimization algorithms, such as [9], and so on. A* algorithm is a classical optimal heuristic search algorithm. Although the algorithm has certain search efficiency, it still can't realize real-time planning. The degree of approximation to the optimal solution depends on the expression of the heuristic function. Genetic algorithm provides a general framework for solving complex problems. However, due to the flight distance of hypersonic vehicle, the area is larger, and the mapping from phenotype to genotype requires a lot of computing resources. In contrast, the algorithm proposed in this paper classifies and screens no fly zones, simplifies the solution, provides the real-time and quick planning by dynamic programming method, and gets the suboptimal solution. The number of waypoints is greatly reduced, and the amount of calculation decreases sharply, so it can meet the real-time demand of guidance.

## VI. Simulation Analysis

The system and aerodynamic parameters of CAV-H are used. Initial height of CAV-H is 40km. Initial velocity is 4000m/s. Initial velocity path angle is 0°, and the yaw angle is -63.43°. The latitude and longitude of the initial point is (0°, 0°), the terminal height is 30km, the terminal latitude and longitude is (16°, 8°). The process constraints can be seen in [10]. The longitude and latitude of the No.1~6 center of the no fly circle are (3°, 1°), (6°, 3.1°), (7°, 2°), (8°, 4.1°), (10°, 4.8°), (12°, 6.6°), with a radius of 50km. The guidance law of document [10] is used for subsection guidance.

From Figure 6, we can see that the aircraft can avoid multiple no fly zones and the trajectory is smooth. Because No.3 no fly zone is class III no fly zone, no need to circumvent. No. 1 no fly zone and No. 6 no fly zone are class II no fly zones. There are No.6 to avoid and No.1 that do not need to circumvent.

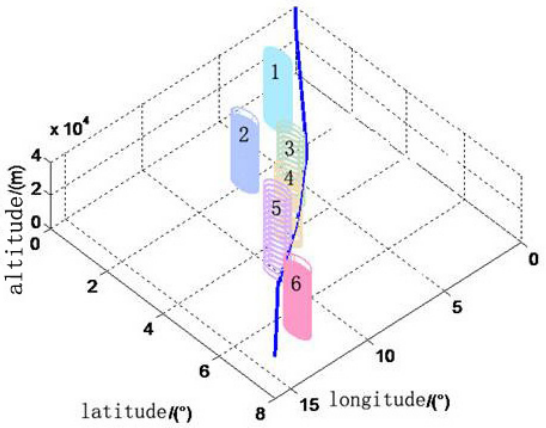


Figure 6. Spatial Trajectory and No Fly Zones Distribution

From figure 7 and Figure 8 shows, at the beginning, large bank angle and attack angle provides normal overload in order to achieve the correct course and satisfy equilibrium; near the No.2, No.4 and No.6 no fly zone, also need bank angle and attack angle to provide the total normal overload to turn; other times the bank angle keep 0 to maintain course, and some attack angle to meet the requirements of equilibrium glide. Because of the control quantity and the total normal overload limited, the three quantities are not beyond the maximum value.

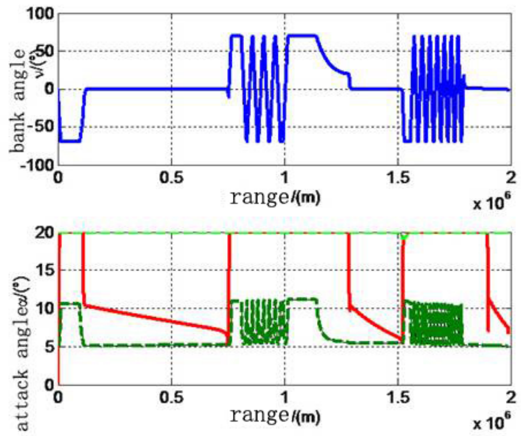


Figure 7. Control Variable Change Over Range

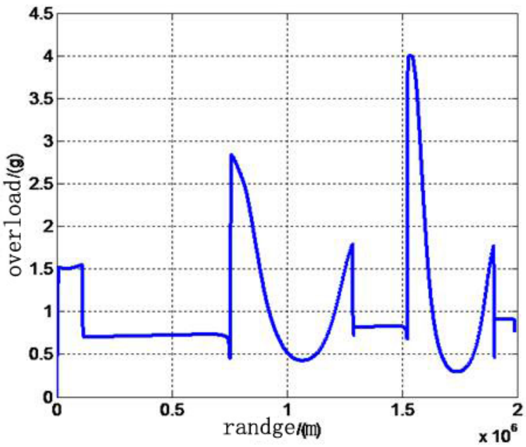


Figure 8. Total Normal Overload Change Over Range

The attack angle corridor is represented by the dotted line in Figure 8. The process constraint is transformed into the attack angle constraint. We can see that the attack angle curve is within two dashed lines, so it can meet the corresponding process constraints. It can be seen from table 1 that the method described in this paper can avoid the avoidance of multiple flight zones, and the terminal latitude and longitude and height error are small at the same time, that is, the terminal constraints are satisfied.

Table 1 terminal state table

| state | Height Error (m) | North Error (m) | East Error(m) | $v_f(m/s)$ |
|---|---|---|---|---|
| value | -6.90 | 57.93 | 118.77 | 1735.09 |

By the above analysis, using the dynamic programming method, the no fly zone constraints translate into waypoint constraints, and then use the quasi equilibrium glide adaptive guidance methods to guide vehicle for each route, so that can achieve multiple no fly zones avoidance and meet a variety of process and terminal constraint.

## VII. Conclusion

Aiming at the problem of gliding guidance with multiple no-fly zones, a novel gliding online no-fly zone avoidance strategy is proposed in this paper. First, the no fly zones are divided into three categories. They are screened by their characteristics. Then, based on the dynamic programming method, the minimum total route offset distance is taken as the decision-making basis, and the avoidance problem of multiple no fly zones is transformed into the waypoint constraint. The flight process is divided into multiple stages by the waypoint got by the avoidance strategy. After that, the multiple no fly zones are avoided by the way of subsection guidance. Finally, the feasibility of the method is verified by simulation analysis. It has a certain reference value for the avoidance and guidance for hypersonic gliding vehicle.

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